Advanced Computer-Aided VLSI System Design
Lab0: Workstation Connection and Practice

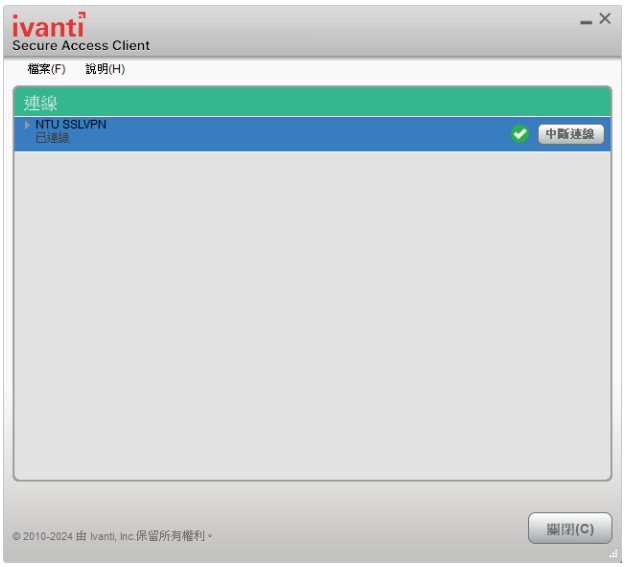
TA: f12943014@ntu.edu.tw

Introduction

In this lab, you will learn how to connect to the ADFP workstation. Also, you will go through how to do a simple project.

VPN

1. First, you need to download the Ivanti Secure Access Client and set up the connection to NTU VPN. You can follow the NTU SSL VPN Service steps to build the VPN connection.
2. Connect the NTU VPN



3. Please make sure you have connected to the NTU VPN while you are using the workstation, including uploading, computing, and summiting your designs, even if you are already under the NTU network IP.

Complete the Lab0

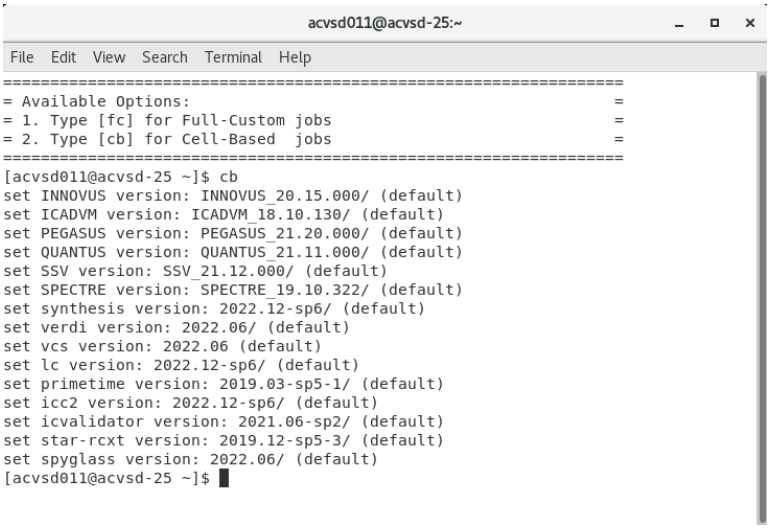
1. Download Lab0 from the NTU cool
2. Unzip the document
   - unzip lab.zip
3. Check out the design of 01_RTL/ADDER.v
4. Run the simulation to on the 231 workstation to make sure the design work properly (optional)

Log In ADFP Workstation and Upload File

1. Please refer to the "ADFP_3.0_connection_method.pdf"

Run the Design Flow

1. Source the CAD tools
   - in terminal: cb



2. Run the rtl simulation
   - cd 01_RTL
   - tcsh ./01_run
3. Run the synthesis
   - cd ../02_SYN
   - tcsh ./01_syn
4. Run the gate simulation
   - cd ../03_GATESIM
   - tcsh ./01_run
   - tcsh ./02_saif_gen
5. Run power analysis
   - cd ../04_power
   - tcsh ./01_power